SEG3502 — Laboratoire 6 — Service Web REST Calculator

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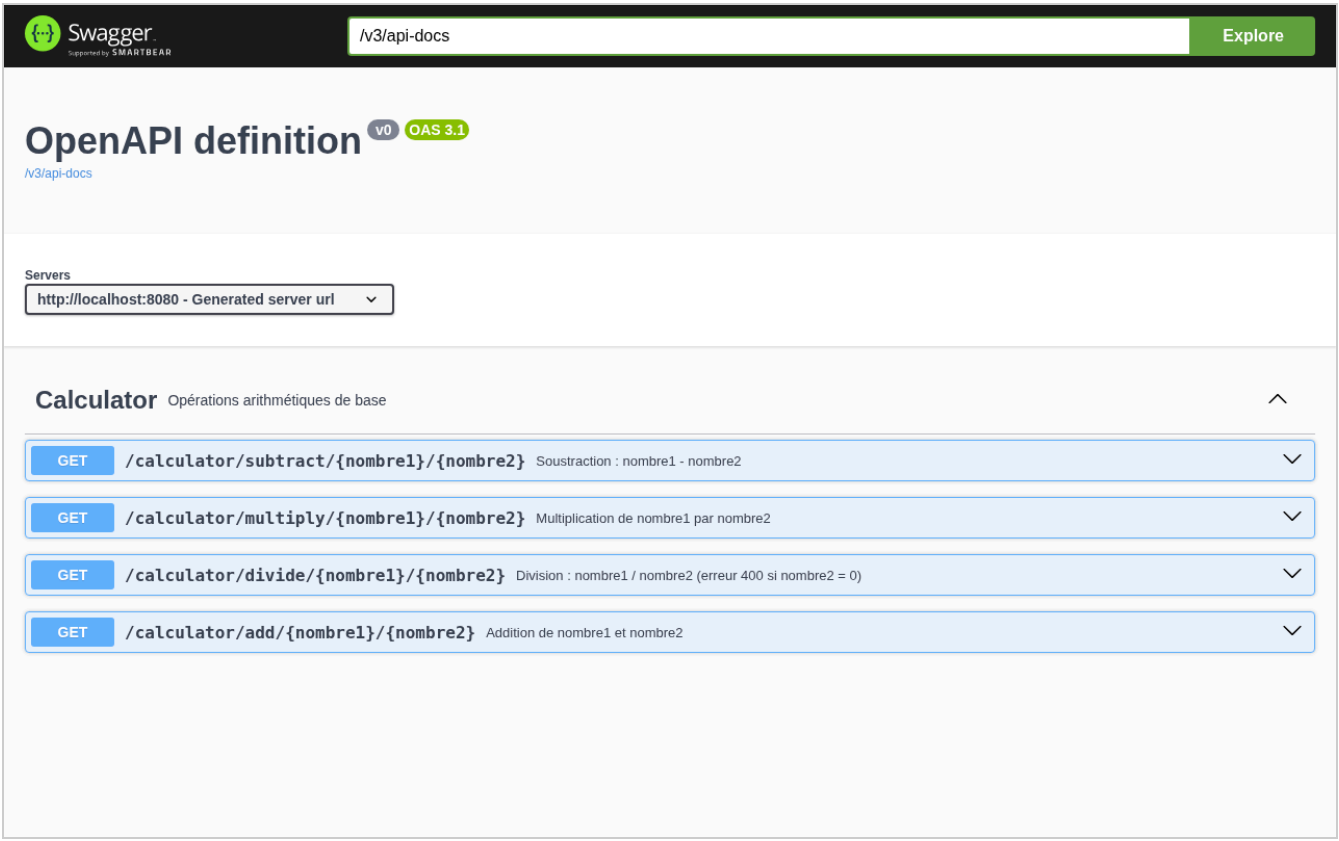
Description du travail réalisé

Développement d'un service Web REST **Calculator** avec Spring Boot 4 et Kotlin (exercice évalué du laboratoire 6). Le contrôleur `CalculatorController` expose quatre opérations GET : `calculator/add`, `subtract`, `multiply` et `divide/{nombre1}/{nombre2}`. Un `@RestControllerAdvice` retourne une erreur 400 (Bad Request) pour une division par zéro ou un paramètre non numérique. La documentation OpenAPI est générée par *springdoc-openapi* (JSON, YAML et Swagger UI). Un test unitaire vérifie les quatre opérations.

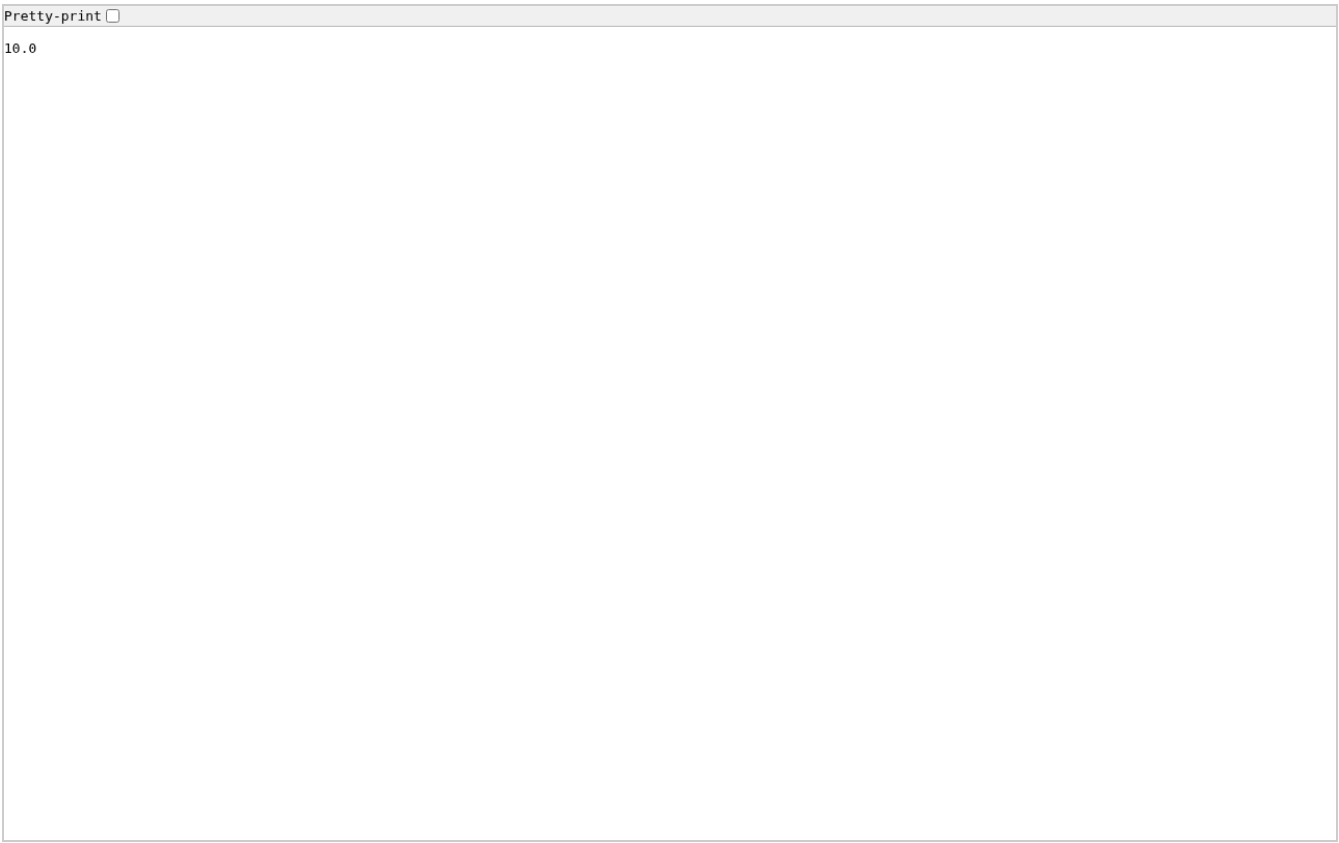
Installation et exécution

1. Installer le JDK 21.
2. Cloner le dépôt puis, à la racine : `./gradlew bootRun` (Windows : `gradlew.bat bootRun`), ou `./gradlew bootJar` puis `java -jar build/libs/calculator-api-0.0.1-SNAPSHOT.jar`.
3. Tester : `curl http://localhost:8080/calculator/add/8/2`.
4. Documentation : `http://localhost:8080/swagger-ui.html` et `http://localhost:8080/v3/api-docs`.
5. Tests : `./gradlew test`.

Captures d'écran



Swagger UI — documentation OpenAPI générée



GET /calculator/add/8/2 → 10.0

Pretty-print

```
4.0
```

GET /calculator/divide/8/2 → 4.0

Unable to process request: Division par zéro

GET /calculator/divide/8/0 → erreur 400

Pretty-print <input type="checkbox" />


```
{"openapi":"3.1.0","info":{"title":"OpenAPI definition","version":"v0"},"servers":[{"url":"http://localhost:8080","description":"Generated server url"}],"tags":
[{"name":"Calculator","description":"Opérations arithmétiques de base"}],"paths":{"/calculator/subtract/{nombre1}/{nombre2}":{"get":{"tags":
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```


/v3/api-docs (JSON)